

Robust deep reinforcement learning for inverter-based volt-var control in partially observable distribution networks[☆]

Qiong Liu^a , Ye Guo^{a,*} , Tong Xu^b

^a Tsinghua Shenzhen International Graduate School, Tsinghua University, Shenzhen, 518071, Guangdong, China

^b Southwest Electric Power Design Institute Co., Ltd. of China Power Engineering Consulting Group, Chengdu, Sichuan, China

HIGHLIGHTS

- We propose a conservative critic to estimate the uncertainty of the state-action value function based on quantile regression technology.
- We propose the surrogate rewards for power loss and voltage violation when the real rewards are unknown.
- The minimal measurement condition of the proposed robust DRL approach is identified.

ARTICLE INFO

Keywords:

Deep reinforcement learning
Volt-var control
Partially observable active distribution network

ABSTRACT

Inverter-based Volt-Var control plays a vital role in regulating voltage and minimizing power loss in active distribution networks (ADNs). However, a key challenge in applying deep reinforcement learning (DRL) to this task lies in the limited measurement deployment of ADNs, which leads to problems of partially observable states and unknown rewards. To address these problems, this paper proposes a robust DRL approach with a conservative critic and a surrogate reward. The conservative critic utilizes the quantile regression technology to estimate a conservative state-action value function based on the partially observable state, which helps to train a robust policy; The surrogate rewards for power loss and voltage violation are designed such that they can be calculated from the limited measurements. The proposed approach optimizes the power loss of the whole network and the voltage profile of buses with measurable voltages while indirectly improving the voltage profile of other buses. Extensive simulations verify the effectiveness of the robust DRL approach under different limited measurement conditions, even when only the active power injection of the root bus and less than 10 % of bus voltages are measurable.

1. Introduction

To achieve a carbon-neutral society, active distribution networks (ADNs) will incorporate more renewable distributed generators (DGs) like PV and wind turbines. The generation of these DGs is highly volatile and uncertain, which can cause problems such as voltage violations and increased power loss. Meanwhile, most DGs are inverter-based (IB) devices that are capable of fast reactive control. This makes it increasingly attractive to use these IB devices in real-time volt-var control (VVC), which minimizes power loss and optimizes voltage profiles in ADNs [1,2]. Such a voltage control paradigm is also known as “IB-VVC” [3].

However, for ADNs with massive renewable DGs, three challenges may emerge in VVC algorithm design: (1) IB-VVC needs to make

real-time decisions to mitigate voltage fluctuations due to the high volatility and uncertainty of renewable energy [4,5]. (2) Current models of ADNs usually involve significant errors or even unknown components [6]. (3) ADNs typically have limited measurement deployments [7].

Model-based IB-VVC approaches have severe limitations in addressing the above challenges. It is a nonlinear programming problem requiring time-consuming iterative steps to converge for off-the-shelf NLP solvers such as IPOPT solvers [6]. To accelerate the convergence process, second-order cone programming [8] or other linearization methods [9,10] have been proposed to reformulate the nonlinear programming problem. The performance of the model-based approach is also vulnerable to parameter errors or unknown models [11]. In addition, when some

[☆] This work was supported in part by the National Key R&D Program of China (2020YFB0906000, 2020YFB0906005).

* Corresponding author.

Email addresses: liuqiong_yl@outlook.com (Q. Liu), guo-ye@sz.tsinghua.edu.cn (Y. Guo), taterx@foxmail.com (T. Xu).

load and generation values are unknown, model-based approaches are difficult to implement. One of the methods is to supplement those unknown data using pseudo-measurements, but such data are not real-time and may be full of noise [12]. State estimation can estimate the unknown data and filter out noise, which has been widely investigated [13]. However, it relies heavily on the accurate power flow model and may not work well for ADNs with inaccurate or unknown models. Sometimes, the pseudo-measurements may not be available for ADNs with weak measurement conditions.

Existing literature has fully demonstrated the advantages of DRL approaches in addressing the former two challenges [3,6,11,14–23]. First, in the application stage, DRL achieves real-time control by performing a fast-forward calculation of policy networks. The time-consuming optimization process is shifted to the training stage [16,24]. Second, DRL is a model-free method that learns to make near-optimal decisions by interacting with ADN environments [6,25–27]. To address the data inefficiency issue of training DRL agents in real ADN environments, a two-stage training framework is proposed where a DRL agent is first pre-trained using a power flow model and then further improved in a real ADN environment [3]. An alternative approach involves employing historical data to train surrogate models as simulation environments for the distribution network, and then DRL agents interact with the surrogate model during training [11,20–23]. However, there is still insufficient research on robust DRL algorithms that are applicable to limited measurement conditions.

DRL is a potential method to address the challenge of limited measurement deployments, but it has seldom been studied. The limited measurements lead to the problem of a partially observable state and unknown reward in the Markov decision process (MDP) in DRL. In cases where the reward is known, some DRL algorithms, such as SAC [28] and PPO [29], can be applied directly to the partially observable tasks with slight modifications [30]. However, the decision performance may be affected due to the missing real-time measurements [4]. For IB-VVC tasks, the pseudo-measurements can be used to supplement the missing items [11], however they may have high forecasting errors. Paper [20] uses a surrogate model trained on historical data to impute unobservable load information, assuming that the unobservable load, while not immediately visible, can be obtained through calculations and retroactive analysis of historical data. However, for ADNs with limited measurement deployment, some loads or voltage data may remain permanently unobservable, which significantly limits the applicability of surrogate-based methods. Additionally, existing research does not fully address the unknown reward problem due to the limited measurement deployment, which is also a critical challenge in the context of this paper.

Partially observable state problems also exist in multi-agent DRL for decentralized VVC tasks, but they differ significantly from the problems studied in this paper. For multi-agent DRL, the partially observable state only occurs in actors, and critics can still observe the global state [4,14,25,31]. This is because decentralized VVC assumes sufficient measurement deployment but has delayed uploading of measured information due to communication limitations.

Given the literature above, we observe two gaps in studying the problem of limited measurement deployment of ADNs for DRL-based VVC tasks: (1) Paper [11] introduces pseudo-measurements to address the problem of partially observable states. However, sometimes, those pseudo-measurements are difficult to acquire in ADNs. Further research is required for DRL to address this problem. (2) The limited measurement deployments also lead to the unknown rewards, which have not been studied.

To address the problems of the partially observable state and the unknown reward, this paper proposes a robust DRL approach, illustrated in Fig. 1. For the problem of the partially observable state, we utilize the conservative critic based on the quantile regression technology to estimate the conservative state-action value function, which helps to train a robust policy. For the problem of the unknown reward, we leverage

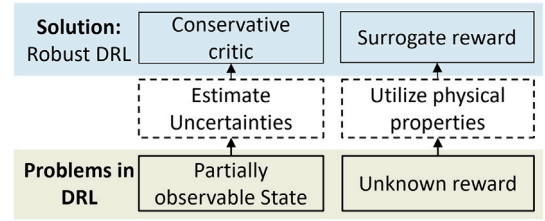


Fig. 1. The overview of the proposed robust deep reinforcement learning approach.

the two rationales of VVC tasks to design the surrogate reward of power loss and voltage violation. *Rationale 1*: For IB-VVC tasks, minimizing the power loss of the entire network is equivalent to minimizing the active power injection of the root node. Therefore, we can design the indirect reward of power loss as the active power injection of the root node. To reflect the power loss change explicitly, we design the surrogate reward, which is the change in the indirect reward of power loss before and after the action execution. *Rationale 2*: In an ADN, the voltages of neighboring buses are coupled. This means that if we optimize the measurable bus voltages, the unmeasurable ones will also be optimized accordingly. Therefore, the surrogate reward of voltage violation is designed as the sum of the voltage violation rates of the measurable nodes. The proposed robust DRL approach works for ADNs with different limited measurement deployments. The minimal measurement condition is that the active power of the root bus and the bus voltages that need to be optimized are measured in real-time. We utilize “robust” to name the proposed robust approach because it addresses the uncertainties led by partial observable states, which is similar to other robust optimization methods addressing the problems of disturbances such as measurement errors or system uncertainties. Compared with the existing DRL-based VVC algorithms, the main technical advancements are summarized as follows:

1. A conservative critic is proposed to estimate the uncertainty of the state-action value function for the partially observable states based on quantile regression technology. Then, a robust policy is proposed by maximizing the value of the conservative critic. Compared with papers [3,14–19,24,32] with the standard critic, the proposed approach with the conservative critic has better performance in reducing voltage violations.
2. The surrogate rewards for power loss and voltage violation are proposed when the real rewards are unknown, which are calculated based on the active power injection of the root bus and the measurable voltage magnitudes. DRL approach with the two surrogate rewards optimizes the power loss of the whole network and the voltage profile of buses with measurable voltage magnitudes, while indirectly improving the voltage profile of other buses.
3. The minimal measurement condition of the proposed robust DRL approach is identified, which is the measurement of the active power injection at the root bus and a few bus voltage magnitudes that need to be optimized. There may be no existing work that has studied the DRL-based VVC approach for ADNs with such limited measurement conditions.

The rest of this paper is organized as follows: Section 2 introduces the preliminaries of IB-VVC and DRL. Section 3 formulates the IB-VVC as a partially observable Markov decision process (POMDP) with two extra items, the partially observable state and the surrogate reward, compared to the existing MDP. Section 4 proposes a robust DRL with the conservative critic and surrogate reward to address the two problems. Simulations and conclusions are provided in Section 5 and 6.

2. Preliminary

2.1. Inverter-based volt-var control

We consider an ADN with $N + 1$ buses, which can be represented by a graph $\mathcal{G}(\mathcal{N}, \mathcal{E})$ with the collection of all buses $\mathcal{N} = 0, \dots, N$ and the collection of all branches $\mathcal{E} = (i, j) \in \mathcal{N} \times \mathcal{N}$. Bus 0 denotes the graph root, and all tree branches are in the direction away from the root.

IB-VVC optimizes power loss and voltage profiles by controlling the reactive power output of distributed generators (DGs) and static Var compensators (SVCs). It can be formulated as an optimal power flow as follows [1,33]:

- (1) The objective function of the optimization problem is minimizing the active power loss P_{loss} of the network:

$$\min P_{loss} = \sum_{i \in \mathcal{N}} P_{Gi} - \sum_{i \in \mathcal{N}} P_{Di} + P_0, \quad (1)$$

where P_{Gi} , P_{Di} are the active powers of DG and load at bus i . For the buses without DG, load, or SVC, the corresponding active and reactive powers are zero. P_0 is the active power injection at the root bus.

- (2) The Dist-flow constraints are:

$$\begin{aligned} P_j &= \sum_{k:j \rightarrow k} P_{jk} - \sum_{i:i \rightarrow j} \left(P_{ij} - \frac{(P_{ij})^2 + (Q_{ij})^2}{(V_i)^2} r_{ij} \right), \\ Q_j &= \sum_{k:j \rightarrow k} Q_{jk} - \sum_{i:i \rightarrow j} \left(Q_{ij} - \frac{(P_{ij})^2 + (Q_{ij})^2}{(V_i)^2} x_{ij} \right), \\ V_j^2 &= V_i^2 - 2(r_{ij}P_{ij} + x_{ij}Q_{ij}) + (r_{ij}^2 + x_{ij}^2) \frac{P_{ij}^2 + Q_{ij}^2}{V_i^2}, \end{aligned} \quad (2)$$

and

$$\begin{aligned} P_j &= \begin{cases} P_0, & j = 0, \\ P_{Gj} - P_{Dj}, & j = 1, \dots, N \end{cases} \\ Q_j &= \begin{cases} Q_0, & j = 0, \\ Q_{Gj} + Q_{Cj} - Q_{Dj}, & j = 1, \dots, N, \end{cases} \end{aligned} \quad (3)$$

where $\forall j \in \mathcal{N}$, P_j , Q_j denote the active and reactive power injections at bus j . P_{jk} , Q_{jk} denote the active and reactive power flow from bus j to k . V_i is the voltage magnitude at bus i . r_{ij} , x_{ij} are the resistance and reactance of branch ij . $i \rightarrow j$ denotes a branch from bus i to bus j . P_{Gj} , Q_{Gj} denote the active and reactive power generations of the DG at bus j . P_{Dj} , Q_{Dj} denote the active and reactive power loads at bus j . Q_{Cj} denotes the reactive power generation of the SVC at bus j . This paper sets the reference voltage bus as $V_0 = 1$ p.u. at bus 0.

- (3) The voltage constraints are:

$$\underline{V}_i \leq V_i \leq \bar{V}_i, \quad (4)$$

where $\bar{V}_i$ and $\underline{V}_i$ are the upper and lower bounds of the voltage magnitude at bus i , respectively.

- (4) The reactive power capability constraints of DGs and SVCs are:

$$\begin{aligned} \underline{Q}_{Gi} &\leq Q_{Gi} \leq \overline{Q}_{Gi}, \\ \underline{Q}_{Ci} &\leq Q_{Ci} \leq \overline{Q}_{Ci}, \end{aligned} \quad (5)$$

where $\overline{Q}_{Gi}$, $\underline{Q}_{Gi}$ and $\overline{Q}_{Ci}$, $\underline{Q}_{Ci}$ are the upper and lower limits of reactive power generation of the DG and SVC at bus i , respectively.

The variables of IB-VVC can be classified as follows: P_D , Q_D , P_G are uncontrollable variables, V , P_0 , Q_0 , I are dependent variables, Q_G , Q_C

are controllable (independent) variables [34]. Decision variables include controllable variables and dependent variables, respectively. Given P_D , Q_D , P_G , IB-VVC minimizes the power loss and ensures the voltage is within its normal range by deciding controllable variables. The dependent variables are solved as well.

2.2. Preliminary of deep reinforcement learning

The objective of the RL algorithm is to learn a policy π to make decisions that maximize the expected discounted accumulated reward from a sequence of MDP data $\mathcal{T}$:

$$\pi^* = \arg \max_{\pi} \mathbb{E}[R(\mathcal{T})], \quad (6)$$

where the policy is

$$a \sim \pi(\cdot | s), \quad (7)$$

the discounted accumulated reward, also named return, is

$$R = \sum_{t=0}^{\infty} \gamma^t r_t. \quad (8)$$

the sequence of MDP data $\mathcal{T} = (s_0, a_0, s_1, a_1, \dots)$ and, $\mathbb{E}$ is a mathematical operation of expect expected value.

For the sake of designing RL algorithms, the state-action value functions are introduced to evaluate the policy π , which is the expected return of an RL policy starting in the state-action pair:

$$Q^{\pi}(s, a) = \mathbb{E}_{a \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, a_0 = a \right]. \quad (9)$$

Considering the control capabilities of inverter-based devices are fast and independent among different decision steps, DRL only needs to maximize the immediate reward [22,31,35]. We can simplify the state-action value function as

$$Q^{\pi}(s, a) = \mathbb{E}_{a \sim \pi} [r \mid s, a]. \quad (10)$$

Then, the objective of RL is to find a policy π to maximize the state-action value function:

$$\pi^* = \arg \max_{a \sim \pi} Q^{\pi}(s, a). \quad (11)$$

2.3. Quantile regression

Quantile regression is a statistical technique that predicts multiple conditional quantiles of a response variable distribution rather than a single mean value as in traditional mean regression [36]. This approach provides a more comprehensive view of the uncertainty and variability in predictions. The τ^{th} quantile of the conditional distribution of the response variable Y given the predictor variable X can be described as

$$f_{Y|X}(\tau) = \inf \{y : F_{Y|X} \geq \tau\}, \quad (12)$$

where τ is the quantile, F is the cumulative distribution function, and $f_{Y|X}(\tau)$ is the τ^{th} conditional quantile value of Y given X .

The τ^{th} quantile function f can be obtained by solving the optimization task:

$$\begin{aligned} f_{Y|X}(\tau) &= \arg \min_f \mathbb{E} \left[(\tau - 1)_{f_{Y|X}(\tau) > Y} (Y - f_{Y|X}(\tau)) \right. \\ &\quad \left. + \tau_{f_{Y|X}(\tau) < Y} (Y - f_{Y|X}(\tau)) \right]. \end{aligned} \quad (13)$$

The quantile regression model predicts multiple quantile values to describe the distribution of response variable Y by setting multiple quantile τ , e.g., 0.1, 0.2, ..., 0.9. Note that $\tau = 0.5$ indicates estimating the median of the response variable, which is the same as the traditional least squares regression.

3. Problem formulation: partially observable markov decision process

The proposed robust DRL approach is model-free and learns to make decisions from limited measurements of ADNs. We consider the following measurement conditions: (1) Voltage magnitude measurements are deployed only on a few buses that need to be optimized; (2) Active and reactive power injection measurements are deployed on the root bus; (3) Branch current magnitude measurements are deployed only on a few distribution branches; (4) The active and reactive power loads and the active power outputs of DGs are deployed only on a few buses.

IB-VVC of an ADN with a limited measurement deployment is formulated as a simplified partially observable Markov decision process (POMDP) $(S, \mathcal{O}, \mathcal{A}, \mathcal{R}, \hat{\mathcal{R}})$, then is solved by the DRL approach. S is the state space, $\mathcal{O}$ is observation space, $\mathcal{A}$ is action space and $\mathcal{R}$ is the reward function. Different from the traditional POMDP $(S, \mathcal{O}, \mathcal{A}, \mathcal{R}, P)$ [30,37] which assumes the real reward $\mathcal{R}$ is known, there is an additional term: "the surrogate reward $\hat{\mathcal{R}}$ ", to represent the real reward when the real reward is unknown. In addition, the proposed simplified POMDP also omits the transition probability function P because the IB-VVC is a single period optimization task, and DRL only needs to maximize the immediate reward and does not need to consider the future state [22,31,35]. This kind of formulation is also named the contextual bandit [38].

At each time step, the state of the ADN environment is $s \in S$, the DRL agent only observes a partial state $s_o \in \mathcal{O}$, and then outputs an action $a \in \mathcal{A}$ based on its policy $\pi : S \rightarrow \mathcal{A}$. After executing the action, the ADN environment transitions to the next state s' with the corresponding real reward r , whereas the DRL agent only observes a partially observable state s'_o and the surrogate reward $\hat{r}$. The DRL agent collects the data tuple $(s_o, a, \hat{r})$ into the data buffer for training the actor and critic networks.

By referring to the IB-VVC formulation introduced in Section 2.1, the state, partially observable state, action, and reward of POMDP are defined as follows:

3.1. State

The state reflects the working condition completely of the ADN, including the uncontrollable, controllable, and dependent variables. $s = (V, P_0, Q_0, I, P_D, Q_D, P_G, Q_G, Q_C)$. In fact, the uncontrollable variables (P_D, Q_D, P_G) are sufficient for IB-VVC decision-making. We add the redundant measurements $(V, P_0, Q_0, I, Q_G, Q_C)$ to reduce the learning difficulties of DRL [39]. Additionally, similar to the state estimation, the redundant information may represent the state robustly by eliminating the measurement errors. Note that the complete state can only be obtained for an ADN with sufficient measurements.

3.2. Observation

For an ADN with limited measurements deployed, the observation is the partially observable state $s_o = (V_o, P_0, Q_0, I_o, P_{D_o}, Q_{D_o}, P_{G_o}, Q_{G_o}, Q_C)$. The subscript "o" indicates the observable measurements. The vector of controllable variables Q_G, Q_C is a linear mapping of the action of the DRL agent and does not need to be measured.

3.3. Action

The action is related to the controllable variables, which are defined as $a = (a_{Q_G}, a_{Q_C})$. For the convenience of constraining the controllable variables, the activation function of the output layer of the actor network is set as "Tanh". Then the action a is always in $(-1, 1)$. The controllable variables Q_G, Q_C are obtained through linear mapping of a from $(-1, 1)$ to their output capability intervals. We can represent it as $u = 0.5(\bar{u} - \underline{u})a + 0.5(\bar{u} + \underline{u})$, where u is the concatenation of Q_G, Q_C , and $\bar{u}, \underline{u}$ are their upper and bottom bounds. The action can also be considered as

the normalized controllable variable u , which may reduce the learning difficulties of neural networks.

3.4. Reward

The reward is to evaluate the optimization performance of the action, which should be calculated based on the P'_0, P'_G, P'_D of the next step. It consists of two sub-terms: the reward for power loss and the reward for voltage violation.

The reward of power loss r_p is defined as:

$$r_p = - \left(P'_0 + \sum_{i \in \mathcal{N}_G} P'_{Gi} - \sum_{i \in \mathcal{N}_D} P'_{Di} \right). \quad (14)$$

The reward of power loss can also be analytically calculated by $r_p = - \sum_{ij \in \mathcal{E}} r_{ij} \frac{P'^2_{ij} + Q'^2_{ij}}{V'^2_{ij}}$, but it is seldom used because it requires model parameters r_{ij} that are difficult to acquire.

The reward of voltage violation r_v is defined as

$$r_v = - \sum_{i \in \mathcal{N}} c_{vi} [\max(V'_i - \bar{V}, 0) + \max(\underline{V} - V'_i, 0)], \quad (15)$$

where c_{vi} is the weighting penalty coefficient of voltage violation. The negative sign is present because DRL aims to maximize the reward, which is converse to the optimization objectives of IB-VVC.

Eqs. (14) and (15) demonstrate that the reward for power loss is calculated based on the active power of the root bus P_0 , the active powers of loads and generations P_D and P_G . Note that although the latter two are not decision variables in the IB-VVC problem (1–5), they may vary in different samples collected at different times. The reward for voltage violation is calculated from all bus voltages V . When all the active power injections and voltages are not measurable, the rewards for power loss and voltage violation become unknown, which limits the application of DRL.

3.5. Surrogate reward

For the unknown reward of power loss, we first propose the indirect reward and then propose the surrogate reward based on the indirect reward, which is calculated based on the P'_0, V'_o of the next step. The indirect reward is the negative value of active power of the root node,

$$\tilde{r}_p = -P'_0. \quad (16)$$

We can also understand that the indirect reward ensures that power loss at unobservable nodes is also optimized from the perspective of model-based IB-VVC. For a given P_D, Q_D, P_G , when model-based IB-VVC decides a controllable reactive power Q_G, Q_C resulting in a minimum power loss, then the active power output of root bus P_0 is also at its minimum value because real power injections of all other nodes are fixed. This means that minimizing the power loss is equal to minimizing the active power injection of the root bus. For the case of the partially observable state, even P_{Gi}, P_{Di} may not be measurable; it can be inferred from the partially observable states [40]. For similar partially observable states, P_{Gi}, P_{Di} tend to be similar. In those cases, if the DRL agent trials an action and obtains a smaller P_0 compared to other actions, it also indicates a smaller power loss.

Noting that P'_0 includes the information of both r_p and $\sum_{i \in \mathcal{N}} (P'_{Di} - P'_{Gi})$ and r_p usually makes up only a small portion of P'_0 . Approximating P'_0 in DRL algorithms may be more challenging than approximating r_p . It tends to result in a slightly larger approximation error in the critic network and degrade the VVC performance. To alleviate the problem, the surrogate reward is designed as

$$\hat{r}_p = \tilde{r}_p - \tilde{r}_p^-. \quad (17)$$

Note that $\tilde{r}_p = -P'_0$ where P'_0 is the active power injection of root bus measured in the next time step, while $\tilde{r}_p^- = -P_0$ is measured in the recent

time step. For a given partially observable state s_o , $\tilde{r}_p^-$ is determined, if the DRL agent outputs an action a that maximizes the surrogate reward $\hat{r}_p$, it also corresponds to the maximum value of r_p , and $\tilde{r}_p$. In addition, the surrogate reward reflects the change in power loss directly, thus providing an apparent reward signal for DRL agents.

For the unknown reward of voltage violation, the corresponding surrogate reward can be calculated using those measurable bus voltages:

$$\hat{r}_v = - \sum_{i=N'} c_{vi} [\max(V_i' - \bar{V}, 0) + \max(\underline{V} - V_i', 0)]. \quad (18)$$

The surrogate reward of voltage violation $\hat{r}_v$ is calculated based on the measurable voltage magnitude V_o' of the next partial observable state. Under the surrogate reward of voltage violation, the objective of DRL is to eliminate voltage violations of measurable buses directly. Since the voltage in one bus has a mutual effect across other buses, voltages of the unmeasurable buses are optimized indirectly.

4. Robust deep reinforcement learning method

The structure of the proposed robust DRL is shown in Fig. 2. In the training stage, a batch of historical data is sampled from the data buffer. The proposed robust DRL approach trains a conservative critic of voltage violation and a critic of power loss using the sampled data. Then, a robust actor is trained to output the action that maximizes the accumulative value of the two critics. In the execution stage, the robust actor receives the partially observable state from the ADN environment and then outputs an action. After executing the action in the ADN environment, the surrogate reward can be calculated based on available measurements. The data tuple, consisting of the partially observable state, surrogate reward, and action, is then uploaded to the data buffer.

In this section, we first propose the conservative critic to address the problem of the partially observable state. Then, we design a practical robust DRL algorithm that integrates the state-of-the-art DRL algorithm with the proposed surrogate reward and conservative critic. Finally, we investigate the minimum measurement condition required for the proposed robust DRL approach.

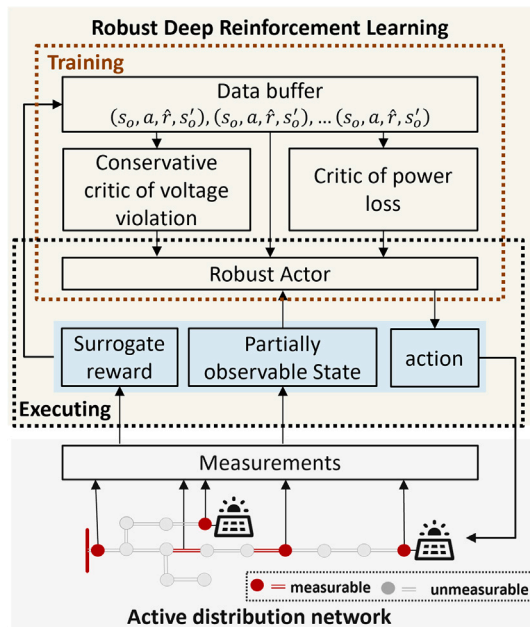


Fig. 2. The framework of robust deep reinforcement learning.

4.1. Conservative critic

In the case of partially observable states, there exist multiple states s corresponding to the same partially observable state s_o . We can use s_o to predict s but with a degree of uncertainty [40].

For the case with partially observable states, the critic network is

$$Q^\pi(s_o, a) = \mathbb{E}_{a \sim \pi} [r | s, a], \quad (19)$$

and the objective of DRL is finding the optimal action a^* ,

$$a^* = \arg \max_a Q(s_o, a). \quad (20)$$

Due to the uncertainties in predicting s_o based on s , $\arg \max_a Q(s, a) \neq \arg \max_a Q(s_o, a)$ and it increases the risk of voltage violations.

There are two methods to address the problem. One method is adding more measurements into the partially observable state s_o to improve the measurement condition, which has been studied in paper [11]. When additional measurements are not available or insufficient, another method proposed in this paper is estimating the uncertainties of reward in the partially observable state, then constructing a conservative critic to estimate the conservative state-action value. The robust policy can be trained with the guidance of the conservative critic network.

Inspired by paper [41], we use quantile regression technology to estimate the uncertainties of state-action value with respect to the partially observable state. Then state-action value (10) is transformed to

$$Q_{r|(s,a)}(\tau) = \inf \{r : F_{r|(s,a)} \geq \tau\}, \quad (21)$$

where τ is the quantile, F is the cumulative distribution function, and $Q_{r|(s,a)}(\tau)$ is the τ^{th} conditional quantile value of r given (s, a) . We set τ as a small value, like 0.2 in this paper, to estimate the conservative state-action value. It can be estimated by solving the optimization problem:

$$Q_{r|(s,a)}(\tau) = \arg \min_Q \mathbb{E} \left[(\tau - 1)_{\hat{Q}(s_o, a) > r} (r - \hat{Q}(s_o, a)) + \tau_{\hat{Q}(s_o, a) < r} (r - \hat{Q}(s_o, a)) \right] \quad (22)$$

To explain why the lower quantile value leads to a conservative state-action value clearly, we plot a schematic diagram of quantile regression as shown in Fig. 3. The lower quantile value, e.g., $\tau = 0.2$ (the green line), assigns a penalty of 0.2 times for data points above the estimated state-action function and a penalty of 0.8 times for data points below it. This asymmetry encourages more data points to fall above the estimated state-action function, thereby leading to a conservative (lower) estimate of the state-action value. $\tau = 0.5$ (the black line) corresponds to the median of the state-action function, which aligns with the standard approach used in existing DRL methods [6,11,18,20,22,25,26,32].

The quantile τ is a tuning parameter to control the level of conservatism in the estimated state-action value. A smaller τ results in a more conservative estimate, which is particularly beneficial for mitigating voltage violations. However, excessively small values of τ may degrade performance in minimizing power loss too much or even destabilize the learning process. Therefore, τ should be carefully tuned based on the

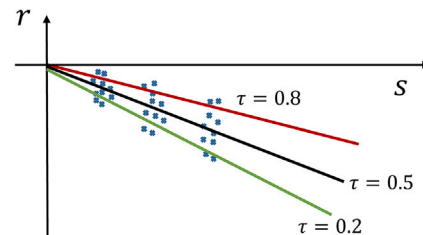


Fig. 3. The schematic diagram of quantile regression.

desired trade-off between voltage violation reduction and power loss minimization. If the performance degradation becomes significant or instability arises, τ can be slightly increased. Empirically, $\tau = 0.2$ is a recommended value in most cases, enhancing voltage violation mitigation with minimal impact on power loss minimization.

In this work, we select quantile regression because it has a simple structure and stable performance, which has been widely adopted in various fields such as statistics, econometrics, and load forecasting [42,43]. Unlike Bayesian neural networks, which require complex sampling techniques like Monte Carlo or variational inference to estimate posterior distributions [44–46], QRNNs utilize a simple quantile regression loss function to train the neural network.

4.2. A practical algorithm: robust soft actor-two-critic

The proposed robust soft actor-two-critic algorithm combines the soft actor-critic (SAC) [28], two-critic DRL framework [47], conservative critic, and surrogate reward. SAC is an off-policy actor-critic deep RL algorithm based on the maximum entropy reinforcement learning framework that provides a robust and sample-efficient learning performance. It is a state-of-the-art DRL algorithm that has been widely utilized in VVC tasks [6,11,25]. The two-critic DRL framework is a specific scheme for IB-VVC, which uses two critic networks to approximate the two objectives of IB-VVC separately [47]. It reduces the learning difficulties of each critic and avoids the mutual effects of the optimization objectives in learning critic networks, thus accelerating the learning process and improving the VVC performance further. In addition, for the proposed robust DRL approach, the two critic networks learning the two objectives separately also provide a convenient way to design the conservative critic solely for voltage violations.

For the IB-VVC task, SAC only learns to maximize the entropy-regularized reward [22,35],

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{a \sim \pi} [r + \alpha H(\pi(\cdot | s_o))], \quad (23)$$

where the entropy is $H(\pi(\cdot | s_o)) = \mathbb{E}_{a \sim \pi(\cdot | s_o)} [-\log \pi(\cdot | s_o)]$. Here, the partially observable state s_o is utilized rather than s .

Integrating the two-critic DRL framework and the surrogate rewards, the critic of power loss $Q_p(s_o, a)$ and the critic of voltage violation $Q_v(s_o, a)$ are introduced:

$$Q_p^\pi(s_o, a) = \mathbb{E}_{a \sim \pi} [\hat{r}_p | s_o, a], \quad (24)$$

$$Q_v^\pi(s_o, a) = \mathbb{E}_{a \sim \pi} [\hat{r}_v | s_o, a]. \quad (25)$$

In this paper, the objective of IB-VVC is to minimize the expected power loss and decrease the risk of voltage violation, so we only apply the conservative critic scheme to the critic of voltage violation. Then, Eq. (25) is transformed to

$$Q_{\hat{r}_v|(s,a)}^\pi(\tau) = \inf \{r : F_{\hat{r}_v|(s,a)}(r) \geq \tau\}, \quad (26)$$

We set τ as a small value to have a conservative critic, like 0.2 in this paper.

The two critics $Q_p(s_o, a)$ and $Q_v(s_o, a)$ are parameterized by two neural networks Q_{ϕ_p} and Q_{ϕ_v} with learnable parameters ϕ_p and ϕ_v . The critic of power loss $Q_{\phi_p}(s_o, a)$ in (24) is learned by minimizing the MSE losses

$$L_{Q_p}(\phi_p) = \frac{1}{|B|} \sum_{(s_o, a, \hat{r}_p) \in B} (Q_{\phi_p}(s_o, a) - \hat{r}_p)^2. \quad (27)$$

The critic of voltage violation $Q_{\phi_v}(s_o, a)$ is learned by minimizing the quantile regression loss

$$L_{Q_v}(\phi_v) = \left[(\tau - 1)_{Q_{\phi_v}(s_o, a) > \hat{r}_v} (\hat{r}_v - Q_{\phi_v}(s_o, a)) + \tau_{Q_{\phi_v}(s_o, a) < \hat{r}_v} (\hat{r}_v - Q_{\phi_v}(s_o, a)) \right]. \quad (28)$$

The actor is a stochastic policy, which is parameterized as

$$\pi_\theta(\cdot | s_o) = \tanh(\mu_\theta(s_o) + \sigma_\theta(s_o) \odot \xi), \quad \xi \sim \mathcal{N}(0, I), \quad (29)$$

where $\mu_\theta, \sigma_\theta$ are neural networks to approximate mean and standard deviations, and ξ is a random variable generated from a standard Gaussian distribution $\mathcal{N}(0, I)$. The actor network $\pi_\theta(\cdot | s)$ learns to maximize the entropy regularized critic value, and the loss function is

$$L_\pi(\theta) = \frac{1}{|B|} \sum_{s_o \in B, a \sim \pi_\theta(\cdot | s_o)} (Q_{\phi_p}(s_o, a) + Q_{\phi_v}(s_o, a) - \alpha \log \pi_\theta(\cdot | s_o)), \quad (30)$$

where $\log \pi_\theta(\cdot | s_o)$ is the entropy of the policy, and α is the temperature parameter. α can be a constant or adjusted by minimizing the loss

$$L(\alpha) = \frac{1}{|B|} \sum_{s_o \in B} (-\alpha \log \pi_\theta(\cdot | s_o) - \alpha H), \quad (31)$$

where H is the target entropy.

The proposed approach optimizes the power loss of the whole network and the voltage profile of buses with measurable voltages while indirectly improving the voltage profile of other buses. The other buses that are closer to the measurable buses would have better voltage optimization performance. The configuration of the measurable buses is decided by the distribution system operator, and voltage sensors would be installed on the pilot nodes or nodes with significant users. The implementation details of the proposed robust soft actor-two-critic algorithm are provided in Algorithm 1.

Deciding the weighting penalty coefficient of voltage violation c_v is crucial for the proposed DRL approach. Since DRL is a model-free approach, we need to tune the c_v to achieve an acceptable performance. A large c_v would decrease the voltage violation rate while increasing the power loss, whereas a small c_v can not eliminate voltage violations successfully.

In our simulation, empirically, we think the value of c_v is suitable when we observe the following two phenomena:

1. During the initial stage, the rate of voltage violations decreases rapidly and approaches zero. This suggests that c_v is sufficiently large to optimize the voltage violations. If the rate of voltage violations cannot approach zero, we need to increase the value of c_v . If c_v is very large, e.g., 1000, and the rate of voltage violations still cannot approach zero, we may need to check whether the VVC problem has feasible solutions.

Algorithm 1 Robust soft actor-two-critic.

Input: Initialize the critic network parameters ϕ_p, ϕ_v , the actor network parameters θ , empty replay buffer D ;

- 1: **for** each environment step **do**
- 2: Execute $a \sim \pi(\cdot | s_o)$ in the ADN environment;
- 3: Observe a reward r_p, r_v and the next state s' ;
- 4: Store $(s, a, \hat{r}_p, \hat{r}_v, s')$ in replay buffer;
- 5: **for** j in range (how many updates) **do**
- 6: Randomly sample a batch of data B from D ;
- 7: Update Q_{ϕ_p} and Q_{ϕ_v} according to

$$\phi_p \leftarrow \phi_p - \lambda_Q \nabla_{\phi_p} L_{Q_p}(\phi_p)$$

$$\phi_v \leftarrow \phi_v - \lambda_Q \nabla_{\phi_v} L_{Q_v}(\phi_v)$$

- 8: Update π_θ according to

$$\theta \leftarrow \theta + \lambda_\pi \nabla_\theta L_\pi(\theta)$$

- 9: **end for**
 - 10: **end for**
-

2. After enough time to learn, about 5 %–10 % of episodes exhibit zero voltage violation, and others have a low voltage violation rate. The low voltage violation rate is caused by the optimization objectives of power loss and voltage violation conflicting in some circumstances. In the training trajectory, DRL algorithms must trail both sides of the voltage boundary many times to approach the optimal solution. If there are larger than 5 %–10 % of episodes exhibiting zero voltage violation, we may need to decrease c_v slightly, and vice versa.

Noting that the topology of distribution networks can change due to reconfigurations, our approach can accommodate these changes by incorporating the topology information into the state s . Specifically, a binary vector can be used to encode the status of circuit breakers that may switch during the control process, and this vector is added to the state s to ensure that the agent is aware of the current network topology. To further enhance learning performance under varying topologies, a multi-task reinforcement learning scheme, as suggested in [18], could be employed. This approach enables joint learning of control policies across multiple tasks, allowing the agent to generalize better to different topological configurations.

When the data available for training DRL are insufficient, we may supplement the training data with the approximate power flow model generated data. In such cases, the proposed DRL approach would be influenced by model quality. To alleviate the problem, we can combine the proposed approach with the two-stage DRL framework [3] that pre-trains a robust DRL agent on an approximate power flow model and then trains the DRL agent online in a real ADN environment. It incorporates the knowledge of approximate models and accelerates the online training process, which makes the proposed method more practical and effective for real-world applications.

Remark 1. We wish to clarify that the two-critic scheme proposed in paper [47] does not contravene the original intent of having two critics in SAC, but is also compatible with SAC [28]. Different from the two-critic scheme [47], the original intent of two critics in SAC is to use the smaller of the two critic values to form the targets in the Bellman error loss functions to address the overestimation of critic values, also known as “clip double Q learning”. We can apply the two-critic scheme [47] to the standard SAC with clipped double Q learning. In this paper, we did not use the clipped double Q learning because IB-VVC is a single-period optimization, and we modified the SAC to maximize the immediate reward rather than the discounted accumulated reward.

4.3. The minimum measurement condition

The proposed robust DRL approach is applicable to different measurement conditions. The minimal measurement condition is the measurement of the active power injection of the root bus P_0 and a few bus voltage magnitudes V_o that need to be optimized. Those measurements can be used as the observation for the proposed DRL algorithm and are also enough to calculate the surrogate reward. Under minimal conditions, DRL algorithms still can work well, which will be verified in the following simulation.

5. Simulation

We conducted numerical simulations on 33, 69, and 118-bus distribution networks to demonstrate the advantages of the proposed robust DRL approach. Those distribution network data were downloaded from Matpower [48] and then converted to Pandapower [49]. The configurations of DGs and SVCs are shown in Table 1. There are 2, 4, and 8 DGs, and 1, 1, and 2 SVCs in 33, 69, and 118-bus, respectively. The reactive power capacities of all DGs and SVCs are set as 3 and 2 MVar. All load power varies between 50 % and 150 % of their default values following uniform distributions. The active generations were a uniform distribution in the interval [0.5, 1.5] MW in the 33 and 69 bus systems and [1, 2] MW in the 118 bus system. The lower and upper bounds of the voltage magnitudes are set as 0.95 and 1.05 p.u., respectively.

Table 1

The configurations of DGs and SVCs in distribution networks.

Network	Devices	buses
33-bus	DG	15, 24
	SVC	32
69-bus	DG	5, 22, 44, 63
	SVC	52
118-bus	DG	33, 44, 50, 53, 76, 97, 106, 111
	SVC	69, 84

Table 2

The configurations of measurements.

O_3	O_2	O_1	Partial state	Complete state
			P_0, V_o	P_0, V
			I_o	I
			$Q_0, P_{D_o}, Q_{D_o}, P_{G_o}$	Q_0, P_D, Q_D, P_G

As shown in Table 2, three measurement conditions were tested to verify the effectiveness of the proposed robust DRL. (1) O_1 : O_1 is the minimal measurement condition where only the active power of the root bus and the voltage magnitudes of a few buses were measurable. Here, we set the voltages in buses with DGs and SVCs to be measurable. (2) O_2 : Measurements in O_1 plus some branch current magnitude measurements. There were 5, 8, and 16 branch current magnitude measurements in 33, 69, and 118 bus distribution networks, respectively. (3) O_3 : Measurements in O_2 plus the active and reactive power of loads in the buses with DGs and SVCs and the active power generations of DGs. The detailed configuration can be seen in our open source code [50]. Three corresponding complete states are also designed. In three complete states, all corresponding voltages, active and reactive power of loads, the active power generations of DGs, and branch currents are measurable.

5.1. Ecust DRL

This section conducts the following simulation to demonstrate the effectiveness of the proposed robust DRL algorithm for ADNs with limited measurement deployments:

1. The standard DRL approach is under the ideal measurement conditions. All measurements including $V, P_0, Q_0, I, P_D, Q_D, P_G, Q_G, Q_C$ are available. This means that the approach utilizes the complete state and complete reward (CC).
2. The proposed robust DRL approach is under three limited measurement conditions. This means that the approach utilizes the partially observable state and surrogate reward (PS). The three simulations are denoted as PS-1, PS-2, and PS-3, respectively.
3. Model-based optimization was formulated based on the AC power flow model and solved by recalling PandaPower [49] with the interior point solver. To be consistent with the DRL approaches, in model-based optimization, the voltage constraints were only added to the buses with voltage measurements. If we assume there is no error in the model, which is indeed the case in our simulations, its results serve as a baseline for the performance of DRL algorithms. Note that the approach only works under the ideal measurement and ideal power flow model conditions.
4. The without-control strategy (W) sets the reactive power generation of DGs and SVC to zero.

All of the simulation experiments are tested on 33, 69, and 118 bus distribution networks to show the scalability of the proposed method. Python was used to implement the algorithms, PyTorch was used for the DRL algorithms, and Pandapower [49] was used to calculate the

Table 3
Parameter settings for the reinforcement learning algorithms.

Parameter	Value
Optimizer	Adam
Activation function	ReLU
Number of hidden layers	2
Actor hidden layer neurons of actor and critic	{512, 512}
Batch size	128
Replay buffer size	3×10^4
Critic learning rate	3×10^{-4}
Actor learning rate	1×10^{-4}
Voltage violation penalty c_{vi}	50
Initial random step	1000
Iterations per time step	4
Entropy target	$-\dim(\mathcal{A})$
Temperature learning rate	3×10^{-4}
Quantile	0.2

balanced power flow for simulating the ADN environments. We trained the DRL agent using 100000 data. The parameter setting for the DRL algorithms is provided in Table 3. In the training process, we tested the DRL algorithms in the same environment at each step.

Fig. 4 presents the simulation results of the standard DRL algorithm with complete state and complete reward (CC), the proposed robust DRL algorithm with three types of partially observable states and surrogate rewards (PS-1, PS-2, PS-3), and the model-based optimization approach with sufficient measurements and an accurate model. In the simulation results, we calculate the reward for power loss using Eq. (14). We calculate the surrogate reward for power loss using Eq. (17). We calculate the reward for voltage violation for all experiments of DRL approaches using Eq. (18). During the experiments, the reward, power loss, and voltage violation were the accumulated values in each episode. Each episode consisted of 1000 steps. The model-based approach also followed the rules of DRL approaches to calculate the reward, power loss, and voltage violation. The voltage violations of the model-based optimization approach were always zero, which were omitted in Fig. 4.

We made three observations from Fig. 4:

First, the model-based approach under the ideal measurements and ideal power flow model achieved the best VVC performance in terms of reward, power loss, and voltage violation. The reward value for the model-based (MB) algorithm exhibits fluctuations because the load and generation profiles are randomly generated and differ in each episode. The power loss changes with the load and generation profiles.

First, the proposed robust DRL under three partially observable conditions (the blue, yellow, and magenta lines) converged close to the standard DRL with ideal measurement conditions (the red line) regarding reward, power loss, and voltage violation, even under the minimal measurement condition, where only 4, 6, and 11 measurements were in 33, 69, and 118 bus distribution networks. The result was consistent among the three distribution networks and the three limited measurement conditions. It indicates that DRL can effectively make decisions by utilizing the partially observable state and surrogate reward, even when only the active power of the root bus and less than 10 % of bus voltages are measurable.

Second, the VVC performance of the robust DRL algorithms improves as the measurement condition increases. By adding a few branch current magnitude measurements based on the minimal measurement condition, the VVC performance and learning efficiency improve, which can be seen by comparing the results of the proposed robust DRL under O1 (the blue line) with the corresponding under O2 (the yellow line). Additionally, measuring reactive and reactive power injections of a few buses further enhances VVC performance and learning efficiency, which can be seen by comparing the results of the proposed robust DRL under O2 (the yellow line) with the corresponding under O3 (the magenta line). These results were apparent across three distribution networks. We can also find that as the number of measurements increases from O1 to O2 to O3, the performance improves, but the improvement rate may decrease.

Third, DRLs with bad observation conditions tend to have unstable learning processes. In the minimal measurement conditions (the blue line), the trajectories of voltage violations in 69 and 118 bus systems fluctuate significantly during the initial learning process. The behavior could be alleviated by generating more data.

Fig. 5 shows the VVC performance of the experiments without control and the proposed robust DRL approach. We made two observations:

First, the proposed DRL approaches under three limited measurement conditions (legend “O1”, “O2”, and “O3”) achieved significantly better performance compared with the experiments without control (legend “W”) in terms of power loss, voltage violation of measurable buses, and of all buses. It demonstrates that the proposed robust DRL for ADNs with limited measurement deployments can optimize the power loss of the whole network and the voltage profile of buses with measurable voltages while indirectly improving the voltage profile of other buses.

Second, the proposed DRL approach was robust to three limited measurement conditions. As the measurement conditions decreased from

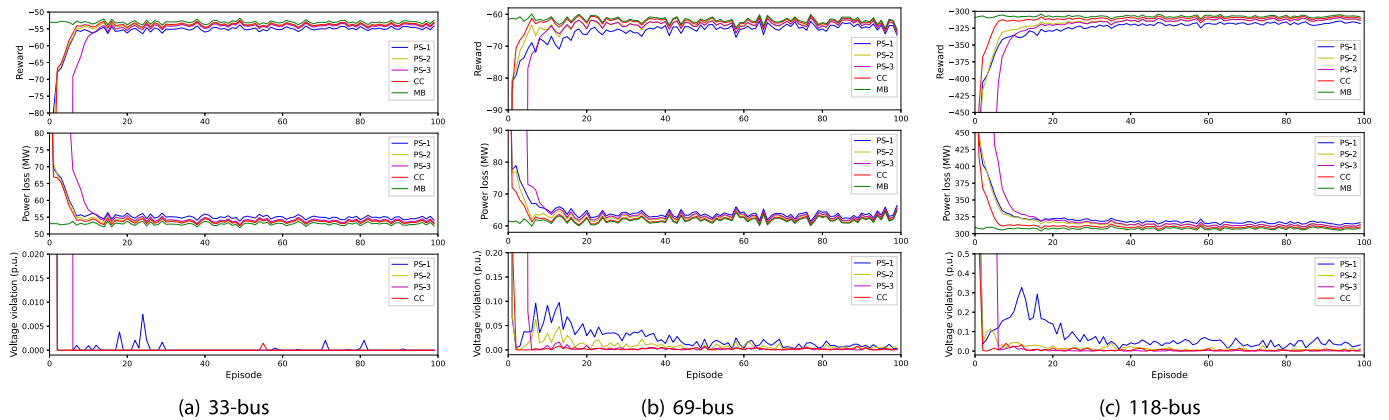


Fig. 4. The testing results of the proposed robust DRL with partially observable state and surrogate reward (PS) under three measurement conditions, the standard DRL with complete state and complete reward (CC), and the model-based optimization (MB). The minimal measurement condition, additional branch currents, and additional power measurements of the few buses and a few branch currents were symbolized as O1, O2, and O3. The results were plotted from the 11th episode. The value of the reward, power loss, and voltage violation is the accumulated value of one episode.

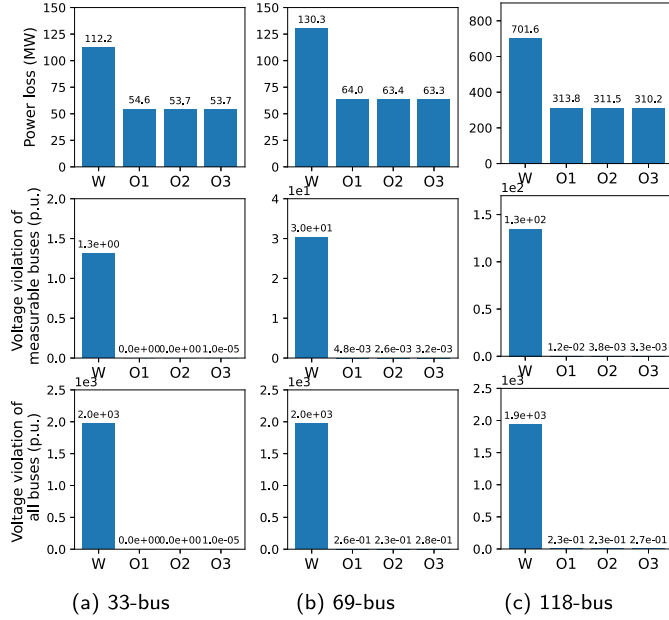


Fig. 5. The VVC performance of without control and the robust DRL robust approach. The figure showed the mean value of the final 20 episodes. "W", "O1", "O2", "O3" represent the experiment without control and robust DRL on three limited measurement conditions ("O1", "O2", "O3").

"O3" to "O2" to "O1", the VVC performance of the proposed DRL only decreased slightly in terms of power loss, voltage violations of measurable buses, and all buses.

5.2. The effect of partially observable state and surrogate reward

To demonstrate the impact of partially observable reward and surrogate reward, we tested the three measurement conditions mentioned above with both surrogate reward and complete reward. Therefore, we designed the four types of simulation settings: (1) complete state and complete reward (CC), (2) complete state and surrogate reward (CS), (3) partially observable state and complete reward (PC), and (4) partially

observable state and surrogate reward (PS). In these DRL approaches, we calculate the reward for voltage violation using Eq. (18).

Comparing simulation results between CC and PC and between CS and PS in Fig. 6 showed the effect of the partially observable state. It was apparent that the partially observable state degraded the VVC performance and slightly increased the power loss and voltage violation rate. We also observed that increasing measurement conditions from O1 to O3 of PC or PS improves VVC performance in power loss and voltage violation. Starting from the minimal measurement condition (O1), VVC performance increased when additional branch current magnitudes were added to the state. Further improvements were achieved by adding additional power injection from a few buses into the state. Generally, the improvement rate from O1 to O2 was larger than that of O2 to O3. There were only two exceptions when comparing the power loss of PC-O2 VS PC-O3, and PS-O2 VS PS-O3 in the 33 bus distribution system, and one exception when comparing the voltage violation of PC-O2 VS PC-O3 in the 118 bus distribution system.

For complete states, adding a few branch currents and additional power injection at a few buses also enhances VVC performance. The results were consistent across all experiments when comparing experiments O1, O2, and O3 between CC and CS. It verified that the redundant information reduces DRL's learning difficulties. Compared with paper [39] which only verified the rule of using bus voltages as redundant information in supervised learning paradigms, this paper also verified the effectiveness of using branch currents as redundant information in DRL paradigms.

Comparing the simulation results of CC vs. CS and PC vs. PS in Fig. 6 shows the effect of the surrogate reward of power loss. We found that the power loss error in CC and PC was slightly higher than the corresponding CS and PS. This verifies that the proposed approach can optimize the power loss of the whole network. Reasonably, due to limited measurement deployments, the VVC performance experiences a slight degradation.

The simulation results in Fig. 5 showed the effect of surrogate reward. We observed that the voltage violation of measurable buses was approaching zero. Although the voltage violation of all buses was slightly higher than that of measurable buses, it was still significantly lower than the voltage violation in the experiments without control. This verified that DRL, with the surrogate reward of voltage violation, optimized the voltage violation of measurable buses directly and of other buses indirectly.

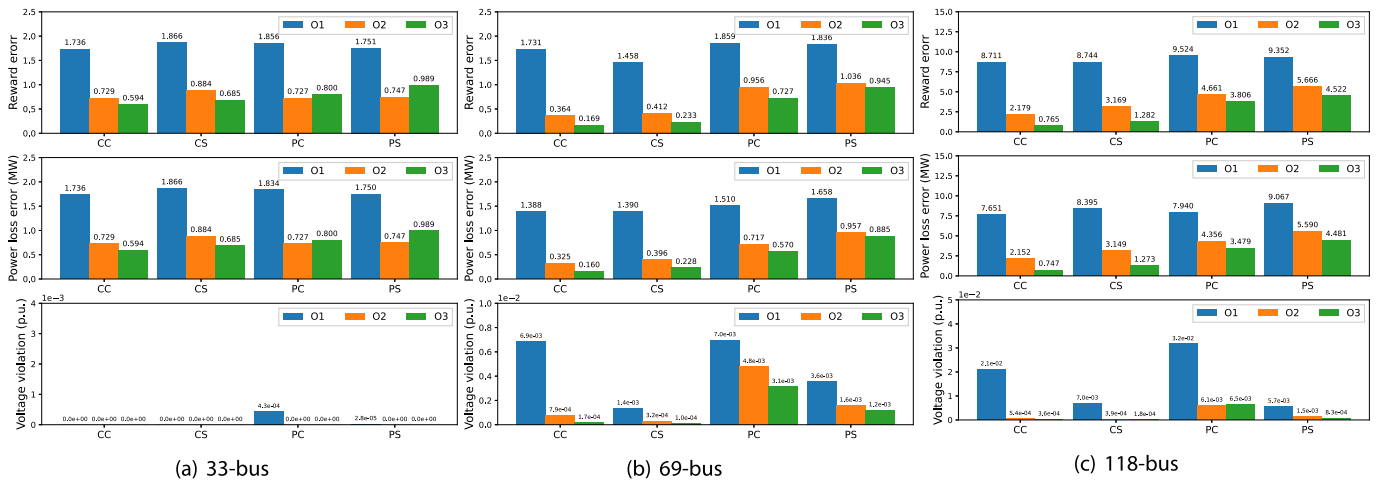


Fig. 6. The optimization error in the final 20 episodes. Optimization error = optimization result of MBO – optimization result of the mentioned method. We test the proposed DRL approach under complete state and complete reward (CC), complete state and surrogate reward (CS), partially observable state and complete reward (PC), and partially observable state and surrogate reward (PS) with three measurement conditions ("O1", "O2", "O3") and three distribution networks.

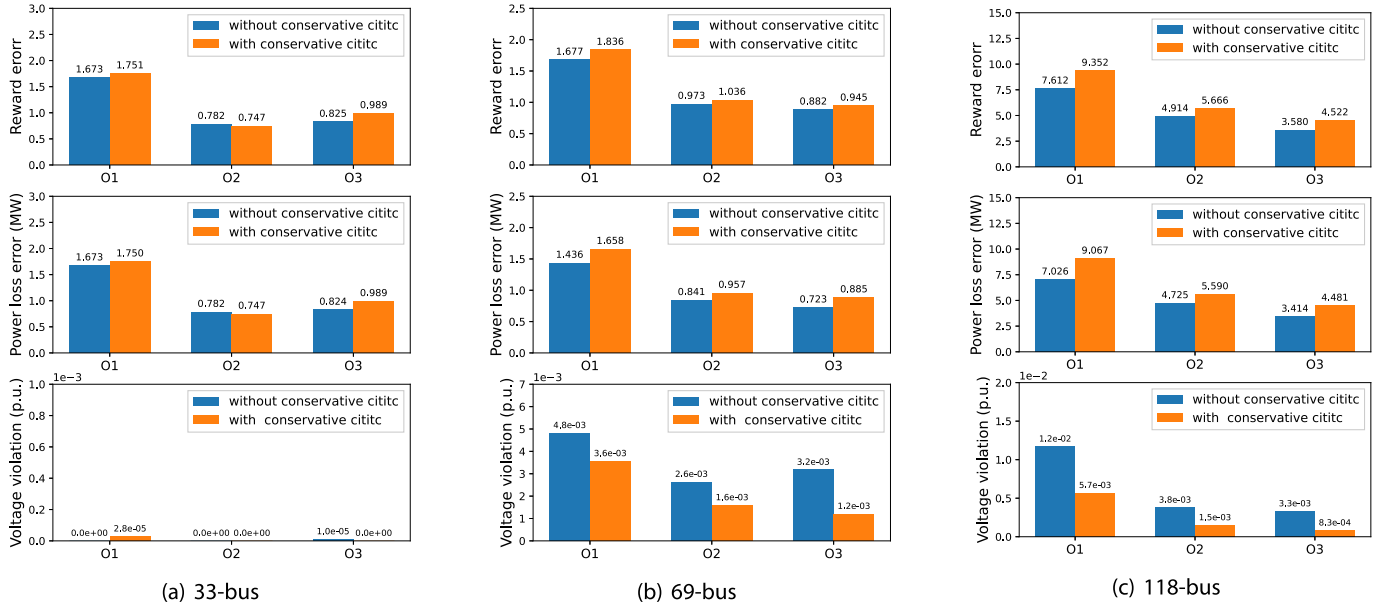


Fig. 7. The optimization error DRL with and without the conservative critic in the final 20 episodes.

5.3. The effect of conservative critic

To demonstrate the effectiveness of the conservative critic, we tested the proposed robust DRL and the standard DRL under three limited measurement deployment conditions. The simulation results are shown in Fig. 7. We observed that the conservative critic reduced the voltage violation value considerably but increased the power loss slightly. The results were consistent when comparing the experiments of DRL with and without the conservative critic under the three limited measurable conditions.

We have conducted a sensitivity analysis on the quantile value in the 118 distribution network. The results are presented in Fig. 8. It shows that as the quantile value increases which means the level of conservatism in the critic decreases, the voltage violation value tends to increase, while the power loss decreases. This trade-off demonstrates

how the choice of τ affects the balance between reducing voltage violations and minimizing power loss.

5.4. Application of the proposed approach for ADNs with topology reconfigurations

To demonstrate the effectiveness of the binary vector encoding scheme in the case of topological reconfiguration, we conducted a simulation on the IEEE 33-bus distribution network. A topology reconfiguration occurred at episode 50, in which the breaker of line (8–9) was opened and the breaker of line (8–14) was closed. Accordingly, the binary vector switched from (1, 0) to (0, 1). Fig. 9 summarizes the simulation results:

1. The reward curve shows only a dip at the moment of topology reconfiguration and quickly converges to a new level, evidencing rapid policy adaptation.

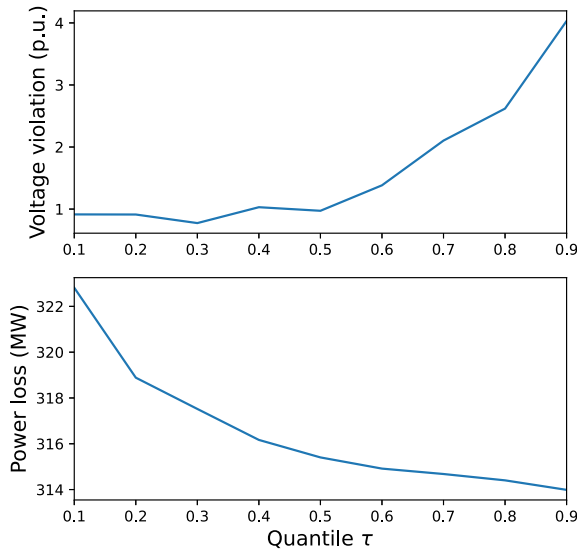


Fig. 8. The change of optimization results of the proposed DRL approach with the increasing of quantile value τ in the 118 distribution network.

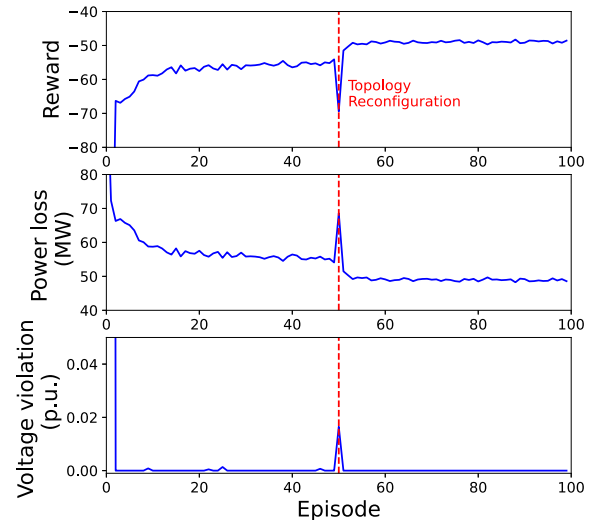


Fig. 9. Controller performance before and after the topology reconfiguration at episode 50.

2. Power loss also has a dip while remaining to converge to a low value in both before and after the change.
3. Although voltage violations momentarily increase due to the topology reconfiguration, they promptly stabilize at low levels.

These results demonstrate that the binary breaker vector enables the DRL agent to learn transferable representations of system topology, allowing it to maintain near-optimal control performance even under dynamic topology reconfigurations with minimal degradation in performance.

5.5. Scalability of the proposed approach for large scale ADNs

To demonstrate the scalability of the proposed method, we evaluated its performance on a 1128-bus distribution network, constructed by combining eight 141-bus systems, as shown in Fig. 10. Each 141-bus network includes 18 PVs and 7 SVCs. The detailed configurations of each 141-bus network are provided in Table 4. In this experiment, we tested the proposed method under a minimal measurement condition, where only the voltage magnitudes of buses equipped with PVs or SVCs and the active and reactive power injections of the slack bus were measurable. Other settings remained consistent with the 118-bus case study.

For such a large network, we increased the hidden layer sizes of both actor and critic networks to (4086, 1024, 512). The batch size was increased to 256, and gradient clipping with a maximum norm value of 10 was applied to improve convergence stability. The penalty coefficient for voltage violation was increased to 100 to provide high penalty signals.

The learning results, presented in Fig. 11, illustrate that the proposed method worked well for the 1128-bus system, with the reward and power loss converging to steady values and the voltage violation approaching zero. The computation time of the proposed robust DRL approach for the 118 and 1128 bus distribution networks is shown in Table 5. After scaling to the 1128 bus distribution network, the decision time for each step is 0.0035 s, which is totally acceptable for real-time applications. These results demonstrate that the proposed approach can successfully learn a Volt-VAR control policy for practical distribution networks with more than 1,000 buses.

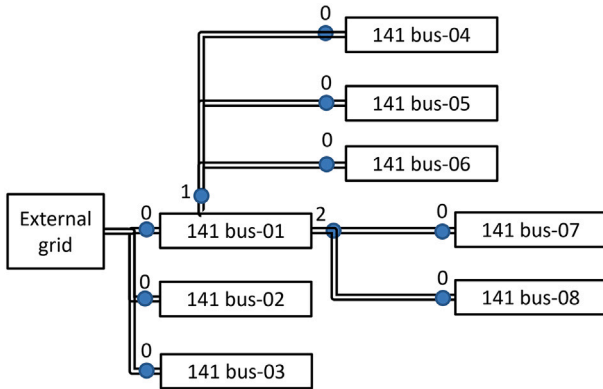


Fig. 10. An 1128-bus distribution network.

Table 4
The configuration of DGs and SVCs in a large distribution network.

Devices	buses
DGs	8, 12, 33, 67, 68, 70, 71, 81, 83, 86, 99, 104, 118, 122, 126, 131, 136, 139
SVC	28, 40, 61, 75, 87, 108, 126

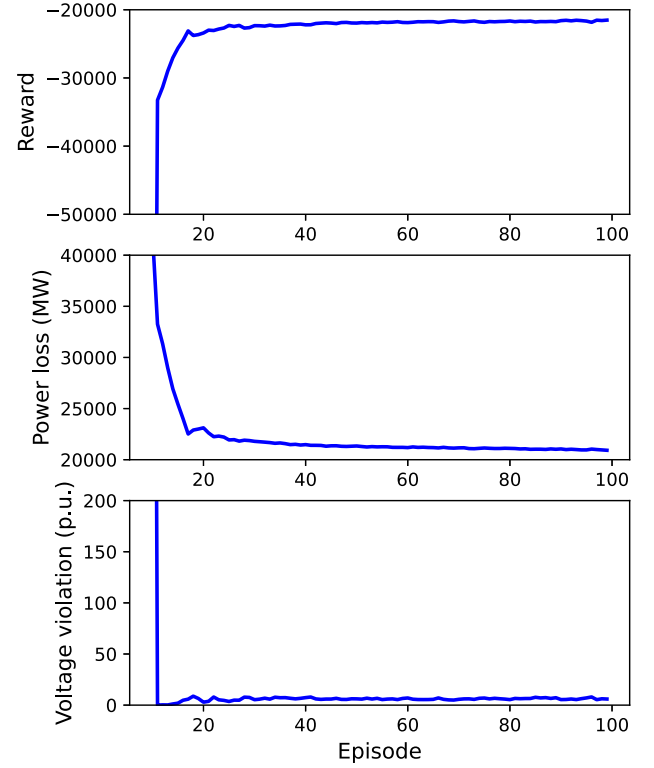


Fig. 11. The learning results of the proposed robust DRL approach on an 1128-bus distribution network.

Table 5
The computation time of the robust DRL approach.

Network	Training time (hour)	Decision time (s)
118-bus	8.5	0.00063
1128-bus	21.1	0.0035

6. Conclusion

This paper proposes a robust DRL approach for IB-VVC in ADNs with limited measurement deployments. Unlike previous works, which utilize pseudo-measurements to improve the measurement condition, the proposed approach directly solves the problem of limited measurement deployments, which led to the problem of the partially observable state and unknown reward of DRL. Then, we propose the conservative critic and surrogate reward to address the two problems separately. Simulation results demonstrated that the scheme operates reliably even under the most restrictive sensing configuration, i.e., only the root-bus active power and a handful of bus voltages.

The framework is well-suited for application in ADNs with limited measurements and uncertain topology characteristics commonly found in modern smart grids. The proposed method can be integrated with existing strategies such as constrained soft actor-critic [6], multi-agent DRL [4], and two- or three-stage DRL architectures [3,35] to further mitigate the effects of partial observability and enhance control robustness. To broaden the applicability of the framework, we plan to extend it in future work to accommodate mixed-integer control problems. This may involve incorporating deep reinforcement learning with mixed-integer programming relaxations, or designing hierarchical architectures that separately handle discrete topology decisions and continuous reactive power control. Such extensions will improve the compatibility of our

method with real-world distribution systems, particularly those incorporating heterogeneous devices such as capacitor banks, on-load tap changers, and energy storage units, all of which require coordinated management of discrete and continuous actions under physical and operational constraints.

CRedit authorship contribution statement

Qiong Liu: Writing – original draft, Software, Methodology, Conceptualization. **Ye Guo:** Writing – review & editing, Conceptualization. **Tong Xu:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix: the theoretical foundation for quantile regression under partial observability

The theoretical foundation for applying quantile regression under partial observability is grounded in probability theory [36,51]. In partially observable distribution networks with limited measurements, partial observations s_o correspond to a conditional probability distribution $p_s(s|s_o)$ over the latent full state s . As a result, even under a fixed policy $a \sim \pi$, the state-action value $Q(s_o, a)$ becomes a random variable due to incomplete information.

Let F_Q denote the cumulative distribution function of $Q(s_o, a)$ with respect to $q(s_o, a)$:

$$F_Q(q(s_o, a)) = \Pr(Q(s_o, a) \leq q(s_o, a)). \quad (32)$$

The τ -th quantile of the distribution $Q(s_o, a)$ is defined as:

$$Q_\tau(s_o, a) = \inf \{q(s_o, a) \in \mathbb{R} \mid F_Q(q(s_o, a)) \geq \tau\}, \quad \tau \in (0, 1). \quad (33)$$

This quantile can be estimated using quantile regression, which minimizes the expected pinball loss [36]:

$$Q_\tau(s_o, a) = \arg \min_{\hat{q}(s_o, a) \in \mathbb{R}} \mathbb{E} [\rho_\tau(Q(s_o, a) - \hat{q}(s_o, a))], \quad (34)$$

where the pinball loss function $\rho_\tau(u)$ is defined as:

$$\rho_\tau(u) = u \cdot (\tau - \mathbb{I}_{u < 0}) = \begin{cases} \tau u, & u \geq 0 \\ (\tau - 1)u, & u < 0 \end{cases}. \quad (35)$$

To demonstrate the theoretical soundness of this formulation, we show that the expected pinball loss is minimized when $\hat{q} = Q_\tau(s_o, a)$, i.e., the true τ -quantile:

$$\begin{aligned} \mathbb{E}[\rho_\tau] &= \int_{-\infty}^{q(s_o, a)} (1 - \tau)(\hat{q}(s_o, a) - q(s_o, a)) dF_Q(q(s_o, a)) \\ &\quad + \int_{q(s_o, a)}^{\infty} \tau(q(s_o, a) - \hat{q}(s_o, a)) dF_Q(q(s_o, a)). \end{aligned} \quad (36)$$

To find the minimizing q , we compute the derivative of the expected loss:

$$\begin{aligned} \frac{d}{dq(s_o, a)} \mathbb{E}[\rho_\tau] &= (1 - \tau) \cdot \int_{-\infty}^{q(s_o, a)} dF_Q(q(s_o, a)) \\ &\quad - \tau \cdot \int_{q(s_o, a)}^{\infty} dF_Q(q(s_o, a)) = F_Q(\hat{q}(s_o, a)) - \tau. \end{aligned} \quad (37)$$

Setting the derivative to zero yields:

$$F_Q(\hat{q}) = \tau \Rightarrow \hat{q} = Q_\tau(s_o, a). \quad (38)$$

This proves that the pinball loss is minimized at the τ -quantile, validating the consistency of quantile regression under partial observability.

Data availability

No data was used for the research described in the article.

References

- [1] Farivar M, Clarke CR, Low SH, Chandy KM. Inverter var control for distribution systems with renewables. In: 2011 IEEE international conference on smart grid communications; 2011. p. 457–62.
- [2] Chen Y, Liu Y, Zhao J, Qiu G, Yin H, Li Z. Physical-assisted multi-agent graph reinforcement learning enabled fast voltage regulation for pv-rich active distribution network. *Appl. Energy* 2023;351:121743.
- [3] Liu H, Wu W. Two-stage deep reinforcement learning for inverter-based volt-var control in active distribution networks. *IEEE Trans Smart Grid* 2021;12(3):2037–47.
- [4] Yan R, Xing Q, Xu Y. Multi-agent safe graph reinforcement learning for pv inverters-based real-time decentralized volt/var control in zoned distribution networks. *IEEE Trans Smart Grid* 2024;15(1):299–311.
- [5] Xu Y, Li Z, Zhang J, Zhang J. Distributed robust control strategy of grid-connected inverters for energy storage systems' state-of-charge balancing. *IEEE Trans Smart Grid* 2018;9(6):5907–17.
- [6] Wang W, Yu N, Gao Y, Shi J. Safe off-policy deep reinforcement learning algorithm for volt-var control in power distribution systems. *IEEE Trans Smart Grid* 2020;11(4):3008–18.
- [7] Xu T, Wu W, Hong Y, Yu J, Zhang F. Data-driven inverter-based volt/var control for partially observable distribution networks. *CSEE J Power Energy Syst* 2023;9(2):548–60.
- [8] Chowdhury MM-U-T, Kamalasadan S. A new second-order cone programming model for voltage control of power distribution system with inverter-based distributed generation. *IEEE Trans Ind Appl* 2021;57(6):6559–67.
- [9] Lin H, Shen X, Guo Y, Ding T, Sun H. A linear distflow model considering line shunts for fast calculation and voltage control of power distribution systems. *Appl. Energy* 2024;357:122467.
- [10] Wang Z, Shen X, Sun H, Wu Q. A practical urban distribution network planning method with geographic information system. *IEEE Trans Power Syst* 2024;39(6):7038–49.
- [11] Cao D, Zhao J, Hu J, Pei Y, Huang Q, Chen Z, et al. Physics-informed graphical representation-enabled deep reinforcement learning for robust distribution system voltage control. *IEEE Trans Smart Grid* 2024;15(1):233–46.
- [12] Azimian B, Biswas RS, Moshagh S, Pal A, Tong L, Dasarath G. State and topology estimation for unobservable distribution systems using deep neural networks. *IEEE Trans Instrum Meas* 2022;71:1–14.
- [13] Primadianto A, Lu C-N. A review on distribution system state estimation. *IEEE Trans Power Syst* 2017;32(5):3875–83.
- [14] Liu H, Zhang C, Chai Q, Meng K, Guo Q, Dong ZY. Robust regional coordination of inverter-based volt/var control via multi-agent deep reinforcement learning. *IEEE Trans Smart Grid* 2021;12(6):5420–33.
- [15] Gao Y, Yu N. Model-augmented safe reinforcement learning for volt-var control in power distribution networks. *Appl. Energy* 2022;313:118762.
- [16] Yi Z, Wang X, Yang C, Yang C, Niu M, Yin W. Real-time sequential security-constrained optimal power flow: a hybrid knowledge-data-driven reinforcement learning approach. *IEEE Trans Power Syst* 2024;39(1):1664–80.
- [17] Sun X, Xu Z, Qiu J, Liu H, Wu H, Tao Y. Optimal volt/var control for unbalanced distribution networks with human-in-the-loop deep reinforcement learning. *IEEE Trans Smart Grid* 2024;15(3):2639–51.
- [18] Pei Y, Zhao J, Yao Y, Ding F. Multi-task reinforcement learning for distribution system voltage control with topology changes. *IEEE Trans Smart Grid* 2023;14(3):2481–84.
- [19] Chen X, Qu G, Tang Y, Low S, Li N. Reinforcement learning for selective key applications in power systems: recent advances and future challenges. *IEEE Trans Smart Grid* 2022;13(4):2935–58.
- [20] Pei Y, Ye K, Zhao J, Yao Y, Su T, Ding F. Visibility-enhanced model-free deep reinforcement learning algorithm for voltage control in realistic distribution systems using smart inverters. *Appl. Energy* 2024;372:123758.
- [21] Hossain RR, Yin T, Du Y, Huang R, Tan J, Yu W, et al. Efficient learning of power grid voltage control strategies via model-based deep reinforcement learning. *Mach Learn* 2024;113:2675–700.
- [22] Cao D, Zhao J, Hu W, Ding F, Huang Q, Chen Z, et al. Data-driven multi-agent deep reinforcement learning for distribution system decentralized voltage control with high penetration of pvs. *IEEE Trans Smart Grid* 2021;12(5):4137–50.
- [23] Liu Q, Guo Y, Deng L, Tang W, Sun H, Huang W. Robust offline deep reinforcement learning for volt-var control in active distribution networks. In: 2021 IEEE 5th conference on energy internet and energy system integration (EI2); 2021. p. 442–48.
- [24] Cao D, Hu W, Zhao J, Huang Q, Chen Z, Blaabjerg F. A multi-agent deep reinforcement learning based voltage regulation using coordinated pv inverters. *IEEE Trans Power Syst* 2020;35(5):4120–23.
- [25] Liu H, Wu W. Online multi-agent reinforcement learning for decentralized inverter-based volt-var control. *IEEE Trans Smart Grid* 2021;12(4):2980–90.
- [26] Liu H, Wu W, Wang Y. Bi-level off-policy reinforcement learning for two-timescale volt/var control in active distribution networks. *IEEE Trans Power Syst* 2023;38(1):385–95.
- [27] Yi Z, Xu Y, Wang X, Gu W, Sun H, Wu Q, et al. An improved two-stage deep reinforcement learning approach for regulation service disaggregation in a virtual power plant. *IEEE Trans Smart Grid* 2022;13(4):2844–58.

- [28] Haarnoja T, Zhou A, Hartikainen K, Tucker G, Ha S, Tan J, et al. Soft actor-critic algorithms and applications. 2019 Jan. arxiv:1812.05905.
- [29] Schulman J, Wolski F, Dhariwal P, Radford A, Klimov O. Proximal policy optimization algorithms. 2017 Jul. arxiv:1707.06347.
- [30] Meng L, Gorbet R, Kulić D. Partial observability during drl for robot control. 2022 Sep. arxiv:2209.04999.
- [31] Sun X, Qiu J. Two-stage volt/var control in active distribution networks with multi-agent deep reinforcement learning method. *IEEE Trans Smart Grid* 2021;12(4):2903–12.
- [32] Cao D, Zhao J, Hu W, Yu N, Ding F, Huang Q, et al. Deep reinforcement learning enabled physical-model-free two-timescale voltage control method for active distribution systems. *IEEE Trans Smart Grid* 2022;13(1):149–65.
- [33] Zhang C, Xu Y. Hierarchically-coordinated voltage/var control of distribution networks using pv inverters. *IEEE Trans Smart Grid* 2020;11(4):2942–53.
- [34] Zhang B, Yan Z. Advanced electric power network analysis. 1st ed. Cengage Learning Asia; 2010.
- [35] Nguyen HT, Choi D-H. Three-stage inverter-based peak shaving and volt-var control in active distribution networks using online safe deep reinforcement learning. *IEEE Trans Smart Grid* 2022;13(4):3266–77.
- [36] Koenker R. Quantile regression. Cambridge; New York: Cambridge University Press; 2005.
- [37] Ghosh D, Rahme J, Kumar A, Zhang A, Adams RP, Levine S. Why generalization in rl is difficult: epistemic pomdps and implicit partial observability. In: Advances in neural information processing systems. Vol. 34. Curran Associates, Inc; 2021. pp. 25502–15.
- [38] Sutton RS, Barto AG. Reinforcement learning. 2nd ed. An Introduction, MIT Press; 2018.
- [39] Lei X, Yang Z, Yu J, Zhao J, Gao Q, Yu H. Data-driven optimal power flow: a physics-informed machine learning approach. *IEEE Trans Power Syst* 2021;36(1):346–54.
- [40] Mestav KR, Luengo-Rozas J, Tong L. Bayesian state estimation for unobservable distribution systems via deep learning. *IEEE Trans Power Syst* 2019;34(6):4910–20. arxiv:1811.02756.
- [41] Dabney W, Rowland M, Bellemare MG, Munos R. Distributional reinforcement learning with quantile regression. In: Proceedings of the thirty-second AAAI conference on artificial intelligence. New Orleans, Louisiana, USA: AAAI Press; 2018. p. 2892–901.
- [42] He Y, Xu Q, Wan J, Yang S. Short-term power load probability density forecasting based on quantile regression neural network and triangle kernel function. *Energy* 2016;114:498–512.
- [43] Zhang W, Quan H, Srinivasan D. An improved quantile regression neural network for probabilistic load forecasting. *IEEE Trans Smart Grid* 2019;10(4):4425–34.
- [44] Jospin LV, Buntine W, Boussaid F, Laga H, Bennamoun M. Hands-on Bayesian neural networks – a tutorial for deep learning users. *IEEE Comput Intell Mag* 2022;17(2):29–48. arxiv:2007.06823.
- [45] Papamarkou T, Hinkle J, Young MT, Womble D. Challenges in Markov chain monte carlo for Bayesian neural networks. *Stat Sci* 2022;37(3):425–42.
- [46] Arbel J, Pitas K, Vladimirova M, Fortuin V. A primer on Bayesian neural networks: review and debates. 2023 Sep. 2309.16314arxiv:2309.16314.
- [47] Liu Q, Guo Y, Deng L, Liu H, Li D, Sun H, et al. Two-critic deep reinforcement learning for inverter-based volt-var control in active distribution networks. In: IEEE transactions on sustainable energy; 2024. p. 1–14.
- [48] Zimmerman RD, Murillo-Sánchez CE, Thomas RJ. Matpower: steady-state operations, planning, and analysis tools for power systems research and education. *IEEE Trans Power Syst* 2011;26(1):12–19.
- [49] Thurner L, Scheidler A, Schäfer F, Menke J-H, Dollichon J, Meier F, et al. Pandapower—an open-source python tool for convenient modeling, analysis, and optimization of electric power systems. *IEEE Trans Power Syst* 2018;33(6):6510–21.
- [50] Qiong L. Ikelq/rdr1_po; Apr 2024. https://github.com/ikelq/RDRL_PO
- [51] Chen X, Alan TK, Zhou Y. Efficient quantile regression analysis with missing observations. *J Am Stat Assoc* 2015;110(510):723–41.